

Design Science Capstone Proposal Prompts

Consider the following when drafting your capstone project proposal to ensure you show a connection with the Design Science Program. The capstone project should be a culmination of your degree program and show application of coursework to solve a real-world problem.

Note: You do not need to answer all the questions in the capstone proposal.

1. Coursework Integration & Knowledge Application

- How have design science methodologies and principles learned throughout your coursework influenced the development of your project goals? Provide specific examples of theories, frameworks, or tools you plan to apply.
- Which design science principles or techniques (e.g., design thinking, systems thinking, user-centered design, prototyping, etc.) from your coursework are most relevant to your project, and how will they shape your approach to problem-solving?

2. Project Goals and Relevance

- What specific challenges or opportunities identified in your project will benefit from the application of design science principles? How do these align with current trends or gaps in the field that you've encountered in your coursework?
- How will your design solution aim to create meaningful and sustainable change within the context of your sponsor's needs or industry? How does your academic background support this focus?

3. Proposal Approach and Methodology

- What methods (e.g., qualitative, quantitative, mixed methods) will you employ in your project, and how have you been prepared to apply these techniques through your coursework? How will these methods help you understand and address the design problem effectively?
- Describe how you will integrate interdisciplinary approaches from other courses (e.g., technology, management, user experience) into your project's design process. What cross-disciplinary knowledge will you draw from to inform your design decisions?

4. Design Process and Iteration

- What role does iteration play in your design process, and how have your design science courses equipped you with strategies to refine and improve your solution over time? How will you incorporate feedback loops and testing throughout the project lifecycle?
- How will your coursework in prototyping, usability testing, and evaluation contribute to ensuring your final design solution meets the needs of your sponsor and intended users?

5. Collaboration and Stakeholder Engagement

- In what ways will you apply principles of collaboration, stakeholder analysis, or team dynamics, as learned in your coursework, to engage with your project sponsors, users, and other key stakeholders?
- How will you gather and incorporate stakeholder feedback into the design process? Discuss how your design science training has prepared you to facilitate effective communication and collaboration between different groups.

6. Ethical Considerations and Sustainability

- What ethical considerations are important to your project, and how has your coursework in design ethics or social responsibility shaped your approach to addressing these concerns?
- How does your project aim to ensure long-term sustainability or positive impact on both the environment and society? What strategies from your coursework will you use to ensure your design is sustainable in the long run?

7. Anticipated Outcomes and Impact

- What measurable outcomes or performance indicators will you use to assess the success of your project? How has your coursework equipped you with the tools to measure and evaluate these outcomes?
- How do you expect your project to affect the sponsor's business, organization, or target community? Discuss the potential impact, drawing on what you've learned about design science's role in driving innovation and change.

8. Risk Management and Problem-Solving

- What potential risks or challenges do you foresee during the project, and how have the problem-solving strategies from your coursework prepared you to mitigate these risks?
- How will you balance creativity and feasibility when solving design problems? Discuss how your coursework has helped you approach complex problems from multiple angles.

9. Reflection on Learning and Professional Development

- How has your design science education prepared you to manage a real-world design project from conception through implementation? Reflect on how your academic experiences have developed your skills in project management, design innovation, and leadership.
- How do you envision this capstone project contributing to your personal and professional growth? What insights from your coursework will guide you in navigating this process?

10. Innovation and Future Directions

- In what ways will your project push the boundaries of current design practices or explore innovative solutions? How have your coursework and exposure to emerging trends or technologies influenced your approach to design innovation?
- How will you position your design solution for scalability or adaptability? What frameworks or methods from your coursework will help ensure your project has the potential to evolve over time?